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### ROLLER SKATE

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#### Abstract

A roller skate is provided, wherein the roller skate includes: a boot including a body portion and a sole, the sole having a longitudinal direction and including two receptacles arranged in interval in the longitudinal direction, each receptacle including a wall and at least one slot open on the sole toward the longitudinal direction, the wall defining at least one opening of the at least one slot; a roller assembly including a frame and rollers disposed on the frame, the frame including two insertion portions arranged in interval in the longitudinal direction, the two insertion portions being respectively insertable in and detachable from the two receptacles in the longitudinal direction, each insertion portion being blocked by the wall of one of the two receptacles in a blocking direction away from the sole; and a plurality of fasteners securing the frame to the sole.

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#### Background/Summary

## BACKGROUND OF THE INVENTION

### Field of the Invention

[0001] The present invention relates to a roller skate.

### Description of the Prior Art

[0002] The roller assembly of one kind of conventional roller skate is fastened to the sole of the boot through many fasteners (such as screws). Therefore, it has to form holes on the frame of the roller assembly and the sole of the boot, respectively, for the fasteners to pass through. The structure is complex, difficult to process and manufacture, and inconvenient to assemble and disassemble. The roller assembly of another kind of conventional roller skate includes projections projecting toward the sole of the boot, and the sole of the boot is recessed with slots. After one of the projections is inserted in one of the slots, another one of the projections is then swung to be within another one of the slots and then locked by a locking or latching member. Similarly, the structure is complex, difficult to process and manufacture, and unstable.

[0003] The present invention is, therefore, arisen to obviate or at least mitigate the above-mentioned disadvantages.

## SUMMARY OF THE INVENTION

[0004] The main object of the present invention is to provide a roller skate which has a simple and stable combination structure, and is easy and quick to assemble and disassemble directly from the lateral side.

[0005] To achieve the above and other objects, a roller skate is provided, wherein the roller skate includes: a boot including a body portion and a sole, the sole having a longitudinal direction and including two receptacles arranged in interval in the longitudinal direction, each of the two receptacles including a wall and at least one slot open on the sole toward the longitudinal direction, the wall defining at least one opening of the at least one slot; a roller assembly including a frame and a plurality of rollers disposed on the frame, the frame including two insertion portions arranged in interval in the longitudinal direction, the two insertion portions being respectively insertable in and detachable from the two receptacles in the longitudinal direction, each of the two insertion portions being blocked by the wall of one of the two receptacles in a blocking direction away from the sole; and a plurality of fasteners securing the frame to the sole

[0006] The present invention will become more obvious from the following description when taken in connection with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiment(s) in accordance with the present invention.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a stereogram of an exemplary embodiment of the present invention;

[0008] FIG. 2 is a breakdown drawing of an exemplary embodiment of the present invention;

[0009] FIG. 3 is a cross-sectional view of an exemplary embodiment of the present invention;

[0010] FIG. 4 is a bottom view of a boot according to an exemplary embodiment of the present invention; and

[0011] FIG. 5 is a cross-sectional view of another exemplary embodiment of the present invention.

### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0012] Please refer to FIGS. 1 to 4 for an exemplary embodiment of the present invention. A roller skate 1 of the present invention includes a boot 10, a roller assembly 20 and a plurality of fasteners 30.

[0013] The boot 10 includes a body portion 11 and a sole 12, the sole 12 has a longitudinal direction D1 and includes two receptacles 121 arranged in interval in the longitudinal direction D1, and the two receptacles 121 may be integrally formed on the sole 12 or additionally secured to the

sole **12**. Each of the two receptacles **121** includes a wall **122** and at least one slot **123** open on the sole **12** toward the longitudinal direction **D1**, and the wall **122** defines at least one opening **124** of the at least one slot **123**. The roller assembly **20** includes a frame **21** and a plurality of rollers **22** disposed on the frame **21**, the frame **21** includes two insertion portions **211** arranged in interval in the longitudinal direction **D1**, and the frame **21** may be integrally formed of one piece or a multi-piece structure. The two insertion portions **211** are respectively insertable in and detachable from the two receptacles **121** in the longitudinal direction **D1**, and each of the two insertion portions **211** is blocked by the wall **122** of one of the two receptacles **121** in a blocking direction **D2** away from the sole **12**. The plurality of fasteners **30** secures the frame **21** to the sole **12**. As such, the roller skate **1** has a simple and stable combination structure, and the frame **21** is easy and quick to assemble and disassemble directly from the lateral side.

[0014] The sole **12** further has a width direction **D3**, the wall **122** includes a plurality of blocking flanges **125** arranged in interval in the width direction **D3** and extending in the width direction **D3**, each of the two insertion portions **211** includes a plurality of engaging flanges **212** arranged in interval in the width direction **D3** and extending in the width direction **D3**, and the plurality of engaging flanges **212** are inserted in the at least one slot **123** and blocked by the plurality of blocking flanges **125** in the blocking direction **D2**. The plurality of engaging flanges **212** may be additionally connected to or integrally formed with the frame **21**.

[0015] Specifically, each of the plurality of blocking flanges **125** includes at least one perforation **126** disposed therethrough in the blocking direction **D2**, and each of the plurality of fasteners **30** is disposed through one of the plurality of engaging flanges **212** and one of the at least one perforation **126** and connected to the sole **12**. The at least one the perforation **126** is open on a lateral side of at least one of the plurality of blocking flanges **125** in the width direction **D3**. The sole **12** further includes a plurality of through holes **127**. Each of the plurality of through holes **127** is elongated in the width direction **D3**, which allows adjustment of the roller assembly **20**. A part of the plurality of through holes **127** may be open toward a direction inclined to the blocking direction **D2**, and another part of the plurality of through holes **127** may be open toward a direction parallel to the blocking direction **D2**, which is conducive to the dispersion and transmission of force. The at least one perforation **126** of one of the plurality of blocking flanges **125** and the at least one perforation **126** of another one of the plurality of blocking flanges **125** are of different numbers and disposed offset in the longitudinal direction **D1** relative to each other, which provides greater structural strength and better stability between the boot **10** and the roller assembly **20**, and is conducive to force transmission.

[0016] Preferably, the wall **122** of one of the two receptacles **121** further includes a protrusion **128** between two of the plurality of blocking flanges **125**, and one of the two insertion portions **211** is blocked by the protrusion **128** in the longitudinal direction **D1**, which quickly aligns the plurality of blocking flanges **125** and the plurality of engaging flanges **212**, and firmly and stably secure the frame **21**. The protrusion **128** includes an inclined surface **129**, the inclined surface **129** extends in the width direction **D3** and is inclined to the blocking direction **D2**, and the engaging flange **212** of one of the two insertion portions **211** includes an inclined surface **213** abutted against the inclined surface **129** of the protrusion **128**. During assembling, the inclined surface **129** and the inclined surface **213** can slide relative to each other to make the plurality of blocking flanges **125** and the plurality of engaging flanges **212** more firmly against each other.

[0017] Preferably, a top end of at least one of the plurality of rollers **22** is higher than all bottom surfaces **214** of the plurality of engaging flanges **212**, the sole **12** further includes at least one groove **120**, and at least one of the plurality of rollers **22** is partially within the at least one groove **120** and located between top and bottom sides of the sole **12**, which lowers the center of gravity of the roller skate. In this embodiment, at least one the groove **120** is disposed through the sole **12**, and this allows a relatively larger region of the roller **22** to be accommodated in the groove **120**, thereby lowering the center of gravity of the roller skate and facilitating air circulation. In the

longitudinal direction D1, one of the at least one groove 120 between the two receptacles 121 has a length larger than a radius of each of each of the plurality of rollers 22, which is applicable to roller assemblies that includes different numbers of rollers (FIG. 3 and FIG. 5).

[0018] Although particular embodiments of the invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except as by the appended claims.

## Claims

1. A roller skate including: a boot including a body portion and a sole, the sole having a longitudinal direction and including two receptacles arranged in interval in the longitudinal direction, each of the two receptacles including a wall and at least one slot open on the sole toward the longitudinal direction, the wall defining at least one opening of the at least one slot; a roller assembly including a frame and a plurality of rollers disposed on the frame, the frame including two insertion portions arranged in interval in the longitudinal direction, the two insertion portions being respectively insertable in and detachable from the two receptacles in the longitudinal direction, each of the two insertion portions being blocked by the wall of one of the two receptacles in a blocking direction away from the sole; and a plurality of fasteners securing the frame to the sole.
2. The roller skate of claim 1, wherein the sole further has a width direction, the wall includes a plurality of blocking flanges arranged in interval in the width direction and extending in the width direction, each of the two insertion portions includes a plurality of engaging flanges arranged in interval in the width direction and extending in the width direction, and the plurality of engaging flanges are inserted in the at least one slot and blocked by the plurality of blocking flanges in the blocking direction.
3. The roller skate of claim 2, wherein each of the plurality of blocking flanges includes at least one perforation disposed therethrough in the blocking direction, and each of the plurality of fasteners is disposed through one of the plurality of engaging flanges and one of the at least one perforation and connected to the sole.
4. The roller skate of claim 3, wherein the at least one perforation is open on a lateral side of at least one of the plurality of blocking flanges in the width direction.
5. The roller skate of claim 3, wherein the at least one perforation of one of the plurality of blocking flanges and the at least one perforation of another one of the plurality of blocking flanges are of different numbers and disposed offset in the longitudinal direction relative to each other.
6. The roller skate of claim 1, wherein a top end of at least one of the plurality of rollers is higher than all bottom surfaces of the plurality of engaging flanges.
7. The roller skate of claim 1, wherein the sole further includes at least one groove, and at least one of the plurality of rollers is partially within the at least one groove and located between two sides of the sole.
8. The roller skate of claim 7, wherein one of the at least one groove is disposed through the sole.
9. The roller skate of claim 7, wherein in the longitudinal direction, one of the at least one groove between the two receptacles has a length larger than a radius of each of the plurality of rollers.
10. The roller skate of claim 2, wherein the wall of one of the two receptacles further includes a protrusion between two of the plurality of blocking flanges, and one of the two insertion portions is blocked by the protrusion in the longitudinal direction.
11. The roller skate of claim 10, wherein the protrusion includes an inclined surface, the inclined surface extends in the width direction and is inclined to the blocking direction, and the engaging

flange of one of the two insertion portions includes an inclined surface abutted against the inclined surface of the protrusion.

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